

POLICY ENDOGENEITY: NAIVE DATA FLYWHEEL VS. GOVERNED EPISODE SLICING

Preventing policy collapse from spurious causal correlations via MCU intervention tagging and pre-crash failure buffer purging

✗ NAIVE DATA FLYWHEEL: SPURIOUS CORRELATION & POLICY COLLAPSE

Policy Drifts Toward Wall
 $t = 0 - 3.2 \text{ s}$

Pre-Crash Buffer ⚠
 $t = 3.2 - 4.0 \text{ s}$ (Unsafe)

Human Steers Away
 $t = 4.0 - 6.5 \text{ s}$ (Recovery)

NAIVE RE-TRAINING
Learns: "Drive into wall to succeed"

- **Endogeneity Trap:** Without provenance tags, the model treats the human recovery as consequence of its own pre-crash states.
- **Catastrophic Policy Collapse:** The retrained model actively steers into obstacles to trigger imagined recoveries!

✓ PHYSICAL AI GOVERNED EPISODE SLICING & CRYPTOGRAPHIC LINEAGE

Segment 1: Valid Execution
Truncated at $t = 3.2 \text{ s}$ (Purged)

~~Pre-Crash Buffer~~
✗ PURGED & DISCARDED

Segment 2: Human Recovery
Tagged with SHA-256 Provenance

GOVERNED RE-TRAINING
Learns robust recovery primitives

- **Atomic Hardware Timestamping:** The MCU logs the microsecond timestamp of human torque override.
- **Failure Buffer Purge:** Automatically deletes the preceding 800 ms of trajectory preceding the intervention.
- **Continuous Policy Improvement:** The data flywheel stays healthy, driving asymptotic convergence to zero interventions!